

Sewon Kim

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RESEARCH INTERESTS

Multimodal AI for Autonomous Systems · Large Language Models · Edge Intelligence · Explainable AI

EDUCATION

Jeonbuk National University

Bachelor of Engineering in Computer Science and Engineering; GPA: 4.09/4.5

Jeonju, South Korea

Mar. 2022 – Feb. 2026 (Expected)

- Relevant Coursework: ML/DL, Data Mining, Information Retrieval, Natural Language Processing, Algorithms, Database Systems
- Pursuing micro-degree in Smart Mobility Software (JST Shared University consortium), completion expected Feb. 2026

Universiti Malaya

Exchange Student

Kuala Lumpur, Malaysia

Feb. 2023 – Aug. 2023

RESEARCH EXPERIENCE

Undergraduate Researcher

Natural Language Learning Lab (Advisor: Prof. Hyun-je Song)

Feb. 2025 – Present

Jeonbuk National University

- Developing knowledge graph-enhanced retrieval-augmented generation (RAG) architectures with PyTorch and LangChain to improve multi-hop reasoning in large language models
- Evaluating trade-offs between retrieval accuracy, latency, and context coherence on multi-hop question answering benchmarks using graph-structured memory representations

International Research Scholar (IITP-Funded)

Korean Software Square (KSW) 2025 Summer Program, Purdue University

Jun. 2025 – Aug. 2025

Purdue University

- Designed and field-deployed LoRa-MQTT IoT sensor network for agricultural monitoring, achieving 110m transmission range with sub-second latency across distributed 8-node architecture (per-node cost: \$10–15)
- **Co-authored research manuscript:** “A LoRa and MQTT-Based Monitoring System for Internal and External Beehive Conditions” (8 pages), preparing for conference submission

RESEARCH PROJECTS

AI-Driven Cultural Heritage Damage Detection and Assessment System

Jun. 2025 – Nov. 2025

- Built a cross-platform heritage field-survey app (Flutter + FastAPI + Firebase) that uses a DETR-based model to detect structural damage and output on-site severity grades (A–D) in real-time
- Secured software copyright “Integrated Survey Registration and Damage Diagnosis Management App” (Korea Copyright Commission, Reg. No. C-2025-044615); filed patent on the integrated survey and damage-diagnosis workflow (patent pending)

Hybrid Physical-AI Model for Industrial Emission Impact Prediction

Feb. 2025 – Jun. 2025

- Developed hybrid model integrating AERMOD physical diffusion with Transformer-based deep learning for urban air quality prediction using 7 years of TMS (Tele-Monitoring System) time-series data (2018–2024)
- Built automated multi-source data pipeline with LSTM-based imputation; deployed real-time prediction dashboard with 3D spatial visualization using Streamlit and Folium

YOLOv8-Based Crop Disease Detection for Smart Agriculture

Sep. 2024 – Dec. 2024

- Developed YOLOv8 detection system achieving mAP@0.5 of 0.81 for crop disease identification; built large-scale labeled dataset by combining open-source and field-collected images using RoboFlow
- Improved model performance by transitioning from unsupervised Grounding DINO + SAM to supervised YOLOv8 framework; implemented five-level crop health classification based on disease ratio analysis

TEACHING EXPERIENCE

Teaching Assistant – Algorithms & Linux Programming

Department of Computer Science and Engineering

Sep. 2025 – Present
Jeonbuk National University

- Mentored 80+ students in algorithm design, 50+ students in Linux programming; developed grading scripts and rubrics to standardize homework and exam evaluation
- Held weekly office hours to support students with data structures, algorithmic problem solving, and Unix/Linux environments (compilation, debugging, shell tools)

HONORS & AWARDS

President’s Award for Excellence in Core Competencies (Highest Honor)	2025
President’s Award, Keun-Saram Black Belt Scholarship (Competitive Merit Scholarship, \$3,400)	2025
Excellence Award, Capstone Design Competition (Two-Time Recipient)	2024 – 2025
Commendation, Big Data Supporters Program, Ministry of Science and ICT, Korea	2024
President’s Award for Outstanding Student, Honors Residential College	2024

PROFESSIONAL DEVELOPMENT

International Participant, Data Science & AI Summer School

Bucharest University of Economic Studies (42-hour intensive program)

Jul. 2024
Predeal, Romania

- Selected for competitive international program combining lecture-style instruction and hands-on Python/Jupyter implementation covering LLM reasoning and calibration, knowledge graphs (Neo4j), agentic AI frameworks, time-series forecasting, and ML pipeline orchestration

Data Architecture Semi-Professional (DAsP) Certification

Korea Data Agency

Oct. 2025

LEADERSHIP & SERVICE

Student Representative, Honors Residential College

Jeonbuk National University

2022 – Present

- Organized academic workshops and peer mentoring programs for university residential college students

Lead Student Representative, Big Data Supporters

Big Data Innovation Convergence College, Jeonbuk National University

Mar. 2024 – Mar. 2025

- Led student-driven data literacy initiatives and outreach activities promoting data science education

TECHNICAL SKILLS

Programming: Python, C/C++, Java, SQL, Dart

ML Frameworks: PyTorch, TensorFlow, LangChain, Hugging Face, OpenCV, scikit-learn

Development & Tools: FastAPI, Flask, Flutter, Git, Docker, AWS (EC2, S3), Linux, MQTT

Research Areas: Deep Learning, NLP, Computer Vision, Knowledge Graphs, Time-Series Forecasting, IoT Systems